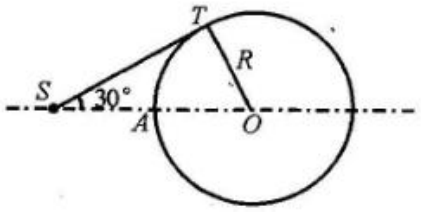
1.

1 T



1

$$\sin 30^\circ = \frac{R}{\overline{SO}}$$

$$\overline{SA} = R$$

A R

A.

$$x \leqslant R \qquad (1)$$

2. $x = R$ α 2

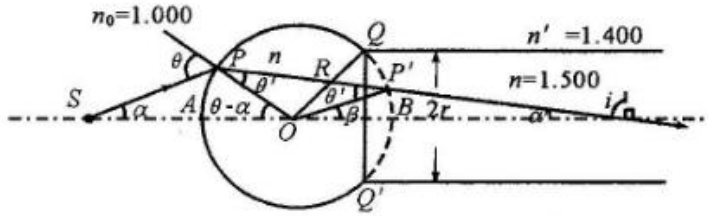
B QBQ' QAQ' θ

θ' P'

β OP' α'

$\triangle OPS$

$$\frac{2R}{\sin\theta} = \frac{R}{\sin\alpha} \qquad (2)$$



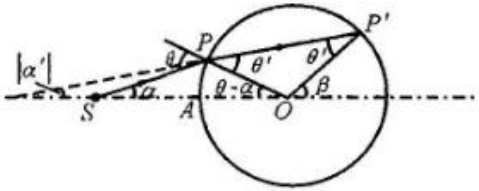
2

$$\sin\theta = 1.5\sin\theta' \qquad (3)$$

$$\beta = 2\theta' - \theta + \alpha \qquad (4)$$

$$\alpha' = \theta - \alpha - \theta' \qquad (5)$$

3



3

A α' (5)

$$|\beta| \leqslant \angle QOB \qquad (6)$$

P'

$$\angle QOB = \arcsin\frac{r}{R}$$